

■ TCY Bot – Base Value Range Strategy

■ Overview

The **TCY Bot** is a high-performance automated Base Value Range (BVR) trading system built with **TypeScript** and deployed on **Synology NAS** using **Node.js** managed by `systemd`. It supports **paper trading**, **Telegram control**, and **auto-restart** with crash recovery.

■ Key Features

■ Core

- Base Value Range strategy loop (60s updates)
- Paper/live calibration via `.env`
- Telegram command control
- Automatic recovery and logging
- Graceful shutdown and restart

■ System Integration

- Runs on Synology NAS (Node.js environment)
- `systemd` service: `tcy-bot-v7.service`
- CLI aliases via `tcybot` for start/stop/status/log

■ Telegram Commands

Command	Description
<code>/status</code>	Show bot status
<code>/log</code>	Show last 15 log lines
<code>/start</code>	Start bot
<code>/stop</code>	Stop bot
<code>/restart</code>	Restart bot
<code>/base</code>	Update base portfolio value
<code>/tcy</code>	Update starting TCY balance
<code>/help</code>	Show available commands

■ Installation & Setup

1 ■ Install Dependencies

```
``bash
cd /volume1/tcy-bot/barracuda-hft
npm install
npm run build:node
``
```

2 ■ Create Systemd Service

File: `/etc/systemd/system/tcy-bot-v7.service`

```
``ini
```

[Unit]

Description=TCY Bot – Base Value Range Strategy (Node.js)

After=network.target

[Service]

ExecStart=/usr/local/bin/npm run start:node

WorkingDirectory=/volume1/tcy-bot/barracuda-hft

Restart=always

User=ssh_admin

Environment=NODE_ENV=production

```
StandardOutput=append:/volume1/tcy-bot/barracuda-hft/logs/tcy-bot-run.log
StandardError=append:/volume1/tcy-bot/barracuda-hft/logs/tcy-bot-run.log
```

```
[Install]
```

```
WantedBy=multi-user.target
```

```
...
```

```
Enable and start:
```

```
```bash
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable tcy-bot-v7.service
```

```
sudo systemctl start tcy-bot-v7.service
```

```
```
```

```
---
```

```
## ■ Alias Commands
```

```
Edit ~/.bashrc`:
```

```
```bash
```

```
alias tcybot="/volume1/tcy-bot/tcybot"
```

```
```
```

```
Then reload:
```

```
```bash
```

```
source ~/.bashrc
```

```
```
```

```
Use:
```

```
```bash
```

```
tcybot start
```

```
tcybot stop
```

```
tcybot status
```

```
tcybot log
```

```
```
```

```
---
```

```
## ■ Telegram Setup
```

```
### 1■■■ Add to `env`:
```

```
```bash
```

```
TELEGRAM_BOT_TOKEN=123456789:ABCdef...
```

```
TELEGRAM_USER_ID=123456789
```

```
```
```

```
### 2■■■ Auto-Notifications
```

```
On startup, you'll receive:
```

```
> ■ TCY Bot connected and ready for commands.
```

```
### 3■■■ Example
```

```
...
```

```
/status → ■ Bot active
```

```
/base 6000 → ■ Updated and restarted
```

```
/tcy 40000 → ■ Updated and restarted
```

```
...
```

```
---
```

■ Monitoring

View logs:

```
```bash
tail -f /volume1/tcy-bot/barracuda-hft/logs/tcy-bot-run.log
```
```

Check service:

```
```bash
systemctl status tcy-bot-v7.service
```
```

Restart manually:

```
```bash
sudo systemctl restart tcy-bot-v7.service
```
```

■ Git Backup

```
```bash
git add .
git commit -m "TCY Bot v7 – Telegram integration"
git push origin production
git push origin v7.0.0
```
```

■ Recommendations

- Secure `.env` and Telegram credentials
- Grant `sudo` access to systemctl for ssh_admin
- Use `npm audit fix` monthly
- Add `/info` command (future): real-time base, TCY, mode

© 2025 Barracuda HFT – TCY Bot Project